

NOTES ON Gravity and Cosmology

version 1.0

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For those who come after

PREFACE

In a sense, you might say I'm getting quite fond of writing these kind of notes.

Not long ago, it occurred to me how cool it is when someone unexpectedly releases a very detailed and all-comprehensive version of their notes, especially when dealing with a course that has a handful of topics often interacting together in unpredictable, yet fascinating, ways.

These notes will be mainly based on *my* own notes of the lectures by Professor Omar Zanusso for the DG, Hamiltonian, and BHs parts, and on the notes of the lectures by Professors Giovanni Marozzi and Daniele Gaggero for the Cosmology part.

These notes refer to the lectures of the academic year 2025-2026, so future versions of these lectures might differ slightly.

At any rate, since I take little to no pride in my messy notes, I'll be often using some of the references you can find on the course catalogue page or that have been cited during class. Either way, I'll do my best keeping track of them all in the bibliography of this humble collection.

You can report errors (whatever their nature might be) and suggestions for additions at g.pannocchia3@studenti.unipi.it or through whatever convoluted way (conventional or not) you prefer¹.

You can find all the notes I've redacted [here](#).

Without further ado, we'd better not lose much more time on a preface and get started with it.

There was Eru, the One, who in Arda is called Ilúvatar; and he made first the Ainur [...] But for a long while they sang only each alone, or but few together, while the rest hearkened; for each comprehended only that part of the mind of Ilúvatar from which he came, and in the understanding of their brethren they grew but slowly. Yet ever as they listened they came to deeper understanding, and increased in unison and harmony.

*Ainulindalë, "The music of the Ainur",
Silmarillion, J. R. R. Tolkien*

¹ I'd like, however, not to see my house stormed by homing pigeons.

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1

BASIS OF DIFFERENTIAL GEOMETRY

Coming soon [1].

2 | HAMILTONIAN APPROACH

Coming soon.

3 | BLACK HOLES

Coming soon.

4 | COSMOLOGY

Coming soon.

Part I

APPENDIX

APPENDIX B: BLACKBODY FUNCTION

Even at an undergraduate level, we're all fairly familiar with *blackbody radiation*. The easiest way to deduce the expression for the energy density of photons in *thermal equilibrium* (STE) inside a cavity is by the means of statistical mechanics.

Remember the Bose-Einstein distribution ($\mu = 0$)

$$n = \frac{1}{\exp(h\nu/kT) - 1}$$

and the phase space density of states (per unit volume)

$$\rho(\nu) d\nu = \frac{4\pi h g \nu^3}{c^3} d\nu$$

from which deducing the expression from internal energy is straightforward. Remembering $g = 2$ is the quantum degeneracy of photons, a simple multiplication of the previous expressions yields

$$U_\nu d\nu = \frac{8\pi h}{c^3} \frac{\nu^3}{\exp(h\nu/kT) - 1} d\nu \quad (1)$$

The energy of a photon is $E_\gamma = h\nu$, meaning the number of photons in the frequency range $\nu \rightarrow \nu + d\nu$ is

$$n_\nu d\nu = \frac{8\pi}{c^3} \frac{\nu^2}{\exp(h\nu/kT) - 1} d\nu \quad (2)$$

which can be integrated over all frequencies, thus obtaining the number density of photons in blackbody radiation

$$n_\gamma = \beta T^3 \quad \beta = \frac{2.4041k^3}{\pi^2 \hbar^3 c^3} \quad (3)$$

Since blackbody radiation is isotropic (it depends only on the absolute temperature T), the definition of mean monochromatic intensity yields

$$B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{\exp(h\nu/kT) - 1} \quad (4)$$

It's important to notice that, in principle, such a fundamental result holds only in *strict thermodynamic equilibrium* (STE), but it's possible to generalize this formulation for less "restrictive" environments.

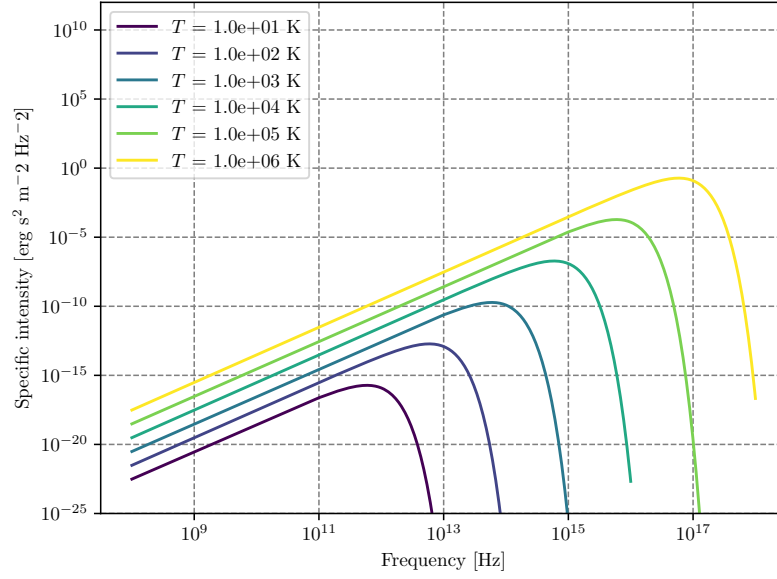


Figure 1: Blackbody frequency spectrum.

An incredible number of important results descends from (4), and it may be worthwhile to cite at least some of them, starting from Stefan-Boltzmann law. We'll use the following result without proving it

$$\int_0^{+\infty} B_\nu(T) d\nu = \frac{2h}{c^2} \frac{\pi^4}{15} \left(\frac{kT}{h} \right)^4$$

Computing the bolometric flux and the bolometric energy density by integrating over all frequencies using what we've just written down, you find the following

$$U(T) = aT^4 \quad F(T) = \sigma_{SB}T^4$$

Clearly the two constants a and σ_{SB} cannot be independent, and are actually related by the integral we've previously calculated. Using for example¹

$$F(T) = \pi \int_0^{+\infty} B_\nu(T) d\nu$$

you can easily find out that the *Stefan-Boltzmann constant* is equal to

$$\sigma_{SB} = \frac{2\pi^5 k^4}{15c^2 h^3}$$

and the relation with a is simply $\sigma_{SB} = ac/4$.

The equation

$$F(T) = \frac{2\pi^5 k^4}{15c^2 h^3} T^4 \tag{5}$$

is what is usually known as the *Stefan-Boltzmann law*.

¹ The emergent flux from an isotropically emitting surface (such as a blackbody) is $\pi \cdot$ brightness which is none other than the specific intensity.

Let us now consider two different regimes for eq.4: $h\nu/kT \ll 1$ and $h\nu/kT \gg 1$. The first yields what is commonly known as the Rayleigh-Jeans Law which is, sadly, pretty much relevant only for radioastronomy.

Since

$$\exp\left(\frac{h\nu}{kT}\right) = 1 + \frac{h\nu}{kT} + o\left(\frac{h\nu}{kT}\right)^2$$

the blackbody radiation assumes the much simpler form of

$$B_\nu^{RJ} = \frac{2\nu^2}{c^2} kT$$

Another important results is achieved in the opposite regime, when the exponential term is rather larger than unity

$$B_\nu^W = \frac{2h\nu^3}{c^2} \exp\left(-\frac{h\nu}{kT}\right)$$

This expression is known as Wien's Law.

Characteristic Temperatures

Starting from the blackbody spectrum we can give various definitions of temperature, that are going to be more and less useful when dealing with stars.

Brightness Temperature

One way of characterizing brightness (specific intensity) at a certain frequency is to give the temperature of the blackbody having the same brightness at that frequency. That is, for any value I_ν we define $T_b(\nu)$ by the relation

$$I_\nu = B_\nu(T_b)$$

this way of specifying brightness has the advantage of being closely connected with the physical properties of the emitter, and has the simple unit (K).

This procedure is used especially in radio astronomy, where the Rayleigh-Jeans law is usually applicable.

$$T_b = \frac{c^2}{2\nu k} I_\nu \quad h\nu/kT \ll 1 \quad (6)$$

Note that the uniqueness of the definition of brightness temperature relies on the monotonicity property of Planck's law. Also note that, in general, the brightness temperature is a function of ν .

Color Temperature

Often a spectrum is measured to have a shape more or less of blackbody form, but not necessarily of the proper absolute value. For example, by measuring F_ν , from an unresolved source we cannot find I_ν , unless we know the distance to the source and its physical size.

By fitting the data to a blackbody curve without regard to vertical scale, a color temperature T_c , is obtained. Often the “fitting” procedure is nothing more than estimating the peak of the spectrum and applying Wien’s displacement law to find a temperature.

Effective Temperature

The effective temperature of a source T_{eff} is derived from the total amount of flux, integrated over all frequencies, radiated at the source. We obtain T_{eff} by equating the actual flux F to the flux of a blackbody at temperature T_{eff}

$$F = \int I_\nu \cos \theta \, d\nu \, d\Omega := \sigma T_{eff}^4 \quad (7)$$

APPENDIX C: CHRISTOFFEL SYMBOLS

For a symmetric, metric-compatible covariant derivative ∇_μ , the Christoffel symbols are given by

$$\Gamma_{\mu\nu}^\alpha = \frac{1}{2}g^{\lambda\alpha}(\partial_\mu g_{\lambda\nu} + \partial_\nu g_{\mu\lambda} - \partial_\lambda g_{\mu\nu}) \quad (8)$$

For the FLWR metric (??), the Christoffel symbols are given by

$$\begin{aligned} \Gamma_{11}^0 &= \frac{a\dot{a}}{1-kr^2} & \Gamma_{11}^1 &= \frac{kr}{1-kr^2} \\ \Gamma_{22}^0 &= a\dot{a}r^2 & \Gamma_{33}^0 &= a\dot{a}r^2 \sin^2 \theta \\ \Gamma_{0j}^i &= \frac{\dot{a}}{a}\delta_j^i = H\delta_j^i & \Gamma_{01}^1 &= \frac{\dot{a}}{a} = H \\ \Gamma_{22}^1 &= -r(1-kr^2) & \Gamma_{33}^1 &= -r(1-kr^2) \sin^2 \theta \\ \Gamma_{12}^2 &= \Gamma_{13}^3 = \frac{1}{r} & \Gamma_{33}^2 &= -\sin \theta \cos \theta \\ \Gamma_{23}^3 &= \cot \theta & \Gamma_{00}^i &= 0 \\ \Gamma_{00}^0 &= 0 & \Gamma_{i0}^0 &= 0 \end{aligned}$$

or related to this by symmetry. The non-zero components of the Ricci tensor are

$$\begin{aligned} R_{00} &= -3\frac{\ddot{a}}{a} \\ R_{11} &= \frac{a\ddot{a} + 2\dot{a}^2 + 2k}{1-kr^2} \\ R_{22} &= r^2(a\ddot{a} + 2\dot{a}^2 + 2k) \\ R_{33} &= r^2(a\ddot{a} + 2\dot{a}^2 + 2k) \sin^2 \theta \end{aligned}$$

and the Ricci scalar is then

$$R = 6 \left[\frac{\ddot{a}}{a} \left(\frac{\dot{a}}{a} \right)^2 + \frac{k}{a^2} \right]$$

ABOUT ME

Hi there, I'm Giacomo, sometimes known as Cael by a very small niche of people, the author of the notes you just finished reading.

I'm a Bachelor-graduate student at the University of Pisa now in the process of undertaking a major (Master degree) in Astronomy and Astrophysics at the same university. In my (meager) free time I'm an avid reader, gamer, anime-enthusiast as well as an independent author in the very process of publishing his first novel.

For those of you that may be interested I'll leave both the link to the novel's page as well to my instagram profile in the QRcodes right below.

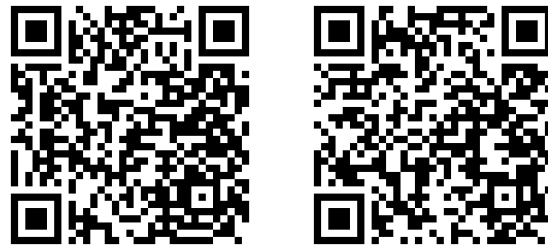


Figure 2: On the left is the link to my instagram profile, on the right the one to my novel's page (Python really has a library for anything, huh?).

Come say hi if you fancy action-fantasy stories with magic, swords, some comedy to garnish and a lot of other stuff :)

At last but not the least, I want to thank all my colleagues and fellow students that helped (and hopefully will help me again) during the drafting of this text, pointing out errors and occasions for further expansion.

If you'd like your names to be included and take your fair share of credits, feel free to contact me and I'll add your name right away in the few lines that are left of this page.

I hope my understading of the complex interplayings going on in our Universe has managed to facilitate your own understading.

If not, well... oopsie.

See you, space cowboys...

Giacomo

BIBLIOGRAPHY

- [1] Sean M. Carroll. *Spacetime and Geometry: An Introduction to General Relativity*. Cambridge University Press, 2019.